

Karim W. Barake

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EDUCATION

Aug 2026 – May 2030 **Boston University Chobanian & Avedisian School of Medicine** Boston, MA

Doctor of Medicine. First-Year Medical Student.

Sep 2022 – May 2026 **Boston University** Boston, MA

Bachelor of Science in Human Physiology. Trustee Scholar, full-merit based scholarship. Member of the Kilachand Honors College, an integrated four-year curriculum. Minor in Business Administration & Management. Undergraduate Research Opportunities Program (UROP) Student Research Award for pursuing a notable scholarly project. Kilachand Internship Program Award for meaningfully focusing on justice work within local communities.

Sargent College Honors Thesis for Distinction – *Small-Molecule BMP4 Agonists as a Potential Anti-Fibrotic Therapeutic Pathway for Proliferative Vitreoretinopathy*. Supervisor: Jeysson Sanchez-Suarez, Ph.D., Research Fellow in Ophthalmology & Dustin Allen, Ph.D., Clinical Assistant Professor.

Kilachand Keystone Project – *School-Based Pediatric Vision Screening Program in Beirut, Lebanon: A Clustered Cross-Sectional Study with Prospective Follow-Up*. Supervisor: Muhammad Zaman, Ph.D., Professor of Biomedical Engineering.

Global Management Experience – Traveled to Beijing and Shanghai as part of a competitive international field study, pre-analyzing and conducting visits to nine companies ranging from multinational corporations such as Lego, Microsoft, and United Energy Group to a neurology virtual reality startup and local manufacturing facilities.

RESEARCH EXPERIENCE

Ophthalmology Department, Mass Eye and Ear at Harvard Medical School | Boston, MA

Student Scientist at the Kim Lab

Leo Kim, M.D., Ph.D., Associate Professor of Ophthalmology Mar 2023 – Present

Lead project investigating small-molecule BMP4 agonist sb4 as a potential therapeutic for proliferative vitreoretinopathy (PVR), designing and executing quantitative cell biology assays, analyzing dose-dependent effects, quantifying molecular pathway activity, and authoring the preliminary manuscript.

Support multiple projects investigating an mRNA-encoded RUNX1-Trap and the Rho-kinase pathway as suppressors of aberrant angiogenesis and key PVR pathological processes, respectively. Conducted cellular assays, performed immunoblotting experiments, and interpreted resulting data.

Direct laboratory image and data analysis using ImageJ, Fiji, GraphPad, and Excel, establishing protocols for human and mouse choroid analysis, migration assays, contractility assays, and choroidal neovascularization quantifications.

Clinical Research Assistant

Mohammad Dahrouj, M.D., Ph.D., Assistant Professor of Ophthalmology Aug 2024 – Dec 2025

Utilized EPIC to review patient charts and analyze surgical outcomes for epiretinal membrane and intraocular lens procedures, examining a broad range of surgical factors to identify patterns associated with both successful outcomes and complications.

Boston University Center on Forced Displacement | Boston, MA

Summer Research Fellow

Aaron Hoshide, Ph.D., Senior Research Fellow May 2025 – Jan 2026

Proposed a biomimicry-inspired policy framework for reforming Lebanon's fragmented primary healthcare system, using the drought-resistant root architecture of grapevines (*Vitis vinifera*) as a structural model to redesign mechanisms for need-based resource sensing, adaptive spatial redistribution of services, and integrated referral networks.

American University of Beirut | Beirut, Lebanon

Research Assistant

Jun 2021 – Aug 2022

Shady Awwad, M.D., Professor of Ophthalmology

Led data collection and statistical analysis of ocular parameters, including angle alpha, angle kappa, and keratometry, from corneal topographies of approximately nine-hundred myopic, hyperopic, and keratoconus eyes, employing retrospective chart review to characterize distribution patterns and identify correlations between these parameters and refractive error, informing preoperative planning for laser and lens-based refractive surgery.

PROJECTS

Ouyouna | Beirut, Lebanon

Sept 2023 – Present

Founder & Executive Director

Founded a non-profit pediatric vision care initiative delivering equitable depoliticized vision screening to public school students in Lebanon. Designed and implemented a task-shifting model training and certifying volunteers and public-school teachers to conduct digital vision screenings using PocDoc, a validated web-based visual acuity platform. Built an end-to-end referral pipeline from school-based screening through free comprehensive exams and spectacle provision.

Screened 791 students across three public schools, identifying 21.6% meeting referral criteria, and provided transportation, free comprehensive eye examinations, and spectacles. Established partnerships with local ophthalmologists and opticians to support the referral pipelines and secured a technology deployment partnership with the National University of Singapore.

Vision for the Homeless | Boston, MA

Aug 2024 – Present

Founder & Executive Director

Built an initiative addressing the vision care gap among Boston's homeless population through direct clinical service, community outreach, and systemic pathway development. Coordinated mobile van outreach and shelter visits across Boston to distribute over-the-counter glasses directly to individuals experiencing homelessness, empowering them to take immediate ownership of their eye health by providing guidance on accessing free comprehensive eye examinations. Established a direct referral pipeline from homeless shelter optometry clinics to local ophthalmology centers to accelerate surgical access and reduce reliance on emergency-stage intervention.

PUBLICATIONS

1.
Gendy, J. E., Bteich, Y., Assaf, J. F., **Barake, K.**, & Awwad, S. T. (2023). Corneal Allogenic Intrastromal Ring Segments as a Therapeutic Method for Intrusion of Synthetic Intrastromal Corneal Ring Segment: A Case Report. *Journal of Refractive Surgery Case Reports*, 3(3). <https://doi.org/10.3928/jrscr-20230814-01>
2.
Bteich, Y., Ibrahim, P., Gendy, J. E., Assaf, J. F., **Barake, K.**, & Awwad, S. T. (2023). Polymethylmethacrylate and Corneal Allogenic Intrastromal Ring Segments: Same Cornea, Different Optical Zones. *Journal of Refractive Surgery Case Reports*, 3(3). <https://doi.org/10.3928/jrscr-20230818-01>
3.
O'Hare, M., Miller, W.P., Arevalo-Alquichire, S., Amarnani, D., Apryani, E., Perez-Corredor, P., Marino, C., Shu, D.Y., Vanderleest, T.E., Muriel-Torres, A., Gordon, H.P., Gunawan, A.L., Kaplan, B.A., **Barake, K.W.**, Bejjani, R.P., Doan, T.H., Lin, R., Delgado-Tirado, S., Gonzalez-Buendia, L., Rossin, E., Zhao, G., Elliott, D., Weinl-Tenbruck, C., Rejman, J., Montrasio, F., Kim, L.A., Arboleda-Vasquez, J.F. (2024). mRNA-encoded RUNX1 Trap: a RunX1 inhibitor with potent antiproliferative effects. *Science Translational Medicine*, 16(775). <https://doi.org/10.1126/scitranslmed.adh0994>
- 4.

- Bteich, Y., Ibrahim, H., **Barake, K.**, Khalil, J., Assaf, J.F., Reinstein, D.Z., Archer, T.J., Awwad, S.T. (2025) Distribution of Angle Alpha and Angle Kappa Chord Magnitude and Axes in Myopic and Hyperopic Refractive Surgery Candidates. *Journal of Refractive Surgery*, 41(6). <https://doi.org/10.3928/1081597X-20250506-04>
5. Zhou, H.W., Ploumi, I., Nodecker, K.N., **Barake, K.**, Gong, D., Davila, J.R., Dahrouj, M., Patel, N.A., Miller, J.B. (2026). Outcomes and Complications of Secondary Intraocular Lens Implantation: Insights from a Vitreoretinal Surgery Cohort. *Ophthalmic Surgery, Lasers and Imaging Retina*, 57(5). <https://doi.org/10.3928/23258160-20260320-01>
 6. Miller, W.P., Arevalo-Alquichire, S., Perez-Corredor, P., Lin, R., Upreti, A., Sanchez-Suarez, J., Vanderleest, T.E., Marino, C., Spencer, D., Scribner, M., Vargas, J., Muriel-Torres, A., Hu, X., Adu-Rutledge, Y., Kaplan, B.A., **Barake, K.W.**, Gunawan, A.L., Vijkumar, A., Shrestha, R., Lennikov, A., Kim, L.A., Arboleda-Vasquez, J.F., Peterson, A., Agrawal, S., Vathipadiekal, V., Loya, C., O'Hare, M. (2026). Development of an Antisense Oligonucleotide Targeting EFEMP1 in Models Relevant to Macular Degeneration. *Molecular Therapy Nucleic Acids*, 37(3). <https://doi.org/10.1016/j.omtn.2026.103024>
 7. Hirsh, T., Miller, W.P., Tsikis, S.T., Arevalo-Alquichire, S., Kaplan, B.A., **Barake, K.W.**, Gunawan, A.L., Vanderleest, T.E., Vijkumar, A., Sanchez-Suarez, J., Scott, F.C., Wang, S.Z., Shei Pan, A., Quigley, M., Kamiya, M., Kim, E.Y., O'Hare, M., Lennikov, A., Kim, L.A., Puder, M. (2026). Inhibition of Run Related Transcription Factor 1 with Ro24-7429 Improves Lung Function in Experimental Models of Idiopathic Pulmonary Fibrosis. *Scientific Reports*, 25073(16). <https://doi.org/10.1038/s41598-026-55553-2>

POSTERS/PRESENTATIONS

1. Awwad, S.T., Bteich Y., Assaf, J.F., **Barake, K.**, Hafezi, F., & Torres, E. Comparative Evaluation of Corneal Haze and Refractive Outcomes After Corneal Cross-Linking Versus Combined Cross-Linking and Selective Wavefront-Guided Transepithelial Photorefractive Keratectomy without Mitomycin-C. Presented at: *European Society of Cornea and Refractive Surgery (ESCRS)*; 2023 Sept 8-12; Vienna, Austria.
2. Bteich, Y., Awwad, S. T., Assaf, J. F., Hafezi, F., **Barake, K.**, & Kozhaya, K. Corneal Haze and Outcomes of Accelerated Corneal Cross-Linking (A-CXL) Versus A-CXL with Customized TransPRK Without Mitomycin-C. Presented at: *American Academy of Ophthalmology (AAO)*; 2023 Nov 3-6; San Francisco, CA.
3. **Barake, K. W.**, Sanchez-Suarez, J., Arevalo-Alquichire, S., Amarnani D., Doobin, D. J., Miller, W. P., Semegran, M., O'Hare, M., & Kim, L. A. Evaluating Netarsudil as a Rho-Kinase Pathway Inhibitor in the Context of Proliferative Vitreoretinopathy. Presented at: *Bay Area Young Investigators Research Conference (BYIRC)*; 2024 Apr 12; Palo Alto, CA.
4. Sanchez-Suarez, J., Arevalo-Alquichire S., Amarnani D., Doobin, D. J., Miller, W. P., **Barake, K. W.**, Semegran, M., O'Hare, M., & Kim, L. A. Rho-kinase Pathway Inhibition for Proliferative Vitreoretinopathy: Assessing the Therapeutic Potential of Netarsudil. Presented at: *Association for Research in Vision and Ophthalmology (ARVO)*; 2024 May 5-9; Seattle, WA.
5. Sanchez-Suarez, J., Bach, B., Upreti, A., Vijkumar, A., Miller, W.P., Adu-Rutledge, Y., **Barake, K. W.**, Lin, R., Semegran, M., Amarnani D., Doobin, D. J., Sohn, E., O'Hare, M., & Kim, L. A. Targeting the Rho-Kinase Pathway for the Treatment of Proliferative Vitreoretinopathy: Netarsudil as a Therapeutic Target. Presented at: *Harvard Ophthalmology's 9th Military Vision Symposium*; 2025; Feb 27-28; Boston, MA.
6. Miller, W.P., Adu-Rutledge, Y., Upreti, A., Vijkumar, A., Sanchez-Suarez, J, **Barake, K. W.**, Kaplan, B. A., Gunawan, A. L., Tsonas, M., ElZaridi, F., O'Hare, M., Chen, F. G., Kim, L. A., Lennikov, A. Characterization of Post-Mortem Human Choroidal Explants as an Ex-Vivo Model of Angiogenesis and Angiofibrosis. Presented at: *Association for Research in Vision and Ophthalmology (ARVO)*; 2025 May 4-8; Salt Lake City, UT.
7. Sanchez-Suarez, J., Upreti, A., Vijkumar, A., Miller, W.P., Adu-Rutledge, Y., **Barake, K. W.**, Lin, R., Semegran, M., Amarnani D., Doobin, D. J., O'Hare, M., & Kim, L. A. Mapping the Cellular Responses to Rho-Kinase Pathway Inhibition in Proliferative Vitreoretinopathy. Presented at: *Association for Research in Vision and Ophthalmology (ARVO)*; 2025 May 4-8; Salt Lake City, UT.
8. Upreti, A., Sanchez-Suarez, J., Miller, W.P., Vijkumar, A., Lin, R., Adu-Rutledge, Y., **Barake, K. W.**, O'Hare, M., Kim, L. A. Characterizing Cellular and Molecular Drivers of Proliferative Vitreoretinopathy Using

Single-Cell RNA Sequencing. Presented at: *Association for Research in Vision and Ophthalmology (ARVO)*; 2025 May 4-8; Salt Lake City, UT.

9.

Upreti, A., Sanchez-Suarez, J., **Barake, K. W.**, Lin, R., Rossin, E. E., O'Hare, M., Kim, L. A. Transcriptomic Landscape of Topoisomerase I Inhibition in Retinal Fibrosis. Presented at: *Harvard Ophthalmology Annual Meeting and Alumni Reunion*; 2025 Jun 12-13; Boston, MA.

10.

Zhou, H.W., Ploumi, I., Nodecker, K., **Barake, K.**, Gong, D.A., Dahrouj, M., Patel, N.A., Miller, J.B. Outcomes and Complications of Secondary Intraocular Lens Implantation. Presented at: *American Society of Retinal Specialists (ASRS)*; 2025 Jul 30 – Aug 2; Long Beach, CA.

11.

Nodecker, K., Baldwin G., **Barake, K.**, Dahrouj, M., Davila, J., Gong, D., Miller, J.B. Comparison of Surgical Outcomes in Epiretinal Membrane (ERM) with the Use of Brilliant Blue versus Indocyanine Green. Presented at: *Women in Ophthalmology*; 2025 Aug 7-10; Amelia Island, FL.

12.

Zhou, H.W., Ploumi, I., Nodecker, K., **Barake, K.**, Gong, D.A., Dahrouj, M., Patel, N.A., Miller, J.B. Outcomes and Complications of Secondary Intraocular Lens Implantation: Insights from a Vitreoretinal Surgery Cohort. Presented at: *The Retina Society*; 2025 Sept 10-13; Chicago IL.

13.

Nodecker, K., Baldwin G., **Barake, K.**, Dahrouj, M., Davila, J., Gong, D., Miller, J.B. Comparison of Surgical Outcomes in Epiretinal Membrane (ERM) with the Use of Brilliant Blue versus Indocyanine Green. Presented at: *American Academy of Ophthalmology (AAO)*; 2025 Oct 18-20; Orlando, FL.

14.

Barake, K.W., Hoshide, A., Samaha, K. Improving Primary Healthcare Access in Lebanon: Biomimicry-Inspired Resilience Using Grapevine (*Vitis vinifera*). Presented at: *Center on Forced Displacement's (CFD) Annual Conference*; 2026 Apr 13-14; Boston, MA.

15.

Sanchez-Suarez, J., Soundararajan, A., Miller, W.P., Lin., R., **Barake, K.W.**, Pattabiraman, P., Kim, L.A. Clusterin as an Endogenous Suppressor of Fibrotic Signaling in Proliferative Vitreoretinopathy. Presented at: *Association for Research in Vision and Ophthalmology (ARVO)*; 2026 May 3-7; Denver, CO. [Accepted]

ARTICLES/TALKS

1.

Barake, K. (2023, July 6). *Revolutionizing Infant Eye Care*. The Ophthalmologist. <https://theophthalmologist.com/subspecialties/revolutionizing-infant-eye-care>

2.

Barake K. (2026, January 14). *The Inequity of Healthcare in Lebanon: Projects Driving Real Change*. AGC Changemaker's Circle, Ali Ghandour Center, Beirut, Lebanon.

SERVICE

Boston University Department of Chemistry | Boston, MA
Principles of Biochemistry (CH373) Peer Mentor Learning Assistant

Jan 2025 - Present

Facilitate weekly discussion sections and host office hours and workshops to support undergraduate students in mastering biochemistry concepts and course material. Directly mentored over 60 students across three semesters, providing individualized feedback and targeted support on challenging topics.

Boston Healthcare for the Homeless | Boston, MA
Optometry Clinical Volunteer at Pine Street Inn

May 2025 – Feb 2026

Conducted baseline eye examinations, including patient history, visual acuity, pupillary assessment, confrontational visual fields, and extraocular mobility prior to dilated examination by the supervising optometrist. Ordered and dispensed spectacles for patients through MassHealth and charitable spectacle-providing programs. Provided real-time medical interpretation for French-, Creole-, and Arabic-speaking patients with limited English proficiency.

United Nations | Bamenda, Cameroon (Remote)
Grant-Writing Volunteer

Oct 2024 – Feb 2025

Identified and completed four grant proposals totaling \$80,000 to fund employment initiatives and social entrepreneurship programs for Cameroonian refugees displaced by internal conflict. Developed a framework for translating refugees' professional expertise and home-country skills into entrepreneurial opportunities within their resettlement communities.

Massachusetts General Hospital | Boston, MA
Eucharistic Minister Volunteer

Jan 2023 – Jun 2023

Visited patient rooms and offered communion to approximately twenty-five Catholic patients per week across all hospital departments. Specifically assigned to Spanish-speaking patients and those in the intensive care unit, providing spiritual companionship through meaningful conversation and prayer.

SKILLS / INTERESTS

Computer: Python, Microsoft Office (Excel, Word, PowerPoint), ImageJ, Fiji, GraphPad, CornealAnalysis, R

Languages: Native/Fluent English, Arabic, French, Beginner Spanish

Interests: Reading, Tennis, Soccer, Swimming, Music, Middle Eastern Studies, Vision Care, Entrepreneurship